A Differentiable Atari VCS

A Complex, Fully Known Ground Truth for Explainable AI

Anonymous Submission

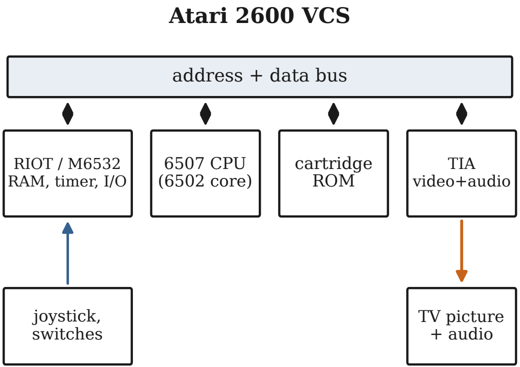
AAAI-27 — Supplementary Video

A real computer architecture, made fully differentiable
Two ports, bit-exact on all 64 Atari games
Gradients with a ground truth

Explanation needs ground truth

- To check an explanation, we must know the true mechanism
- Simple models: mechanism known, but explaining it tests nothing
- Deep networks: explanation needed, but no inner ground truth
- So an explanation can be confident and wrong
- We want a system that is complex, fully known, and differentiable

A study object: the Atari 2600



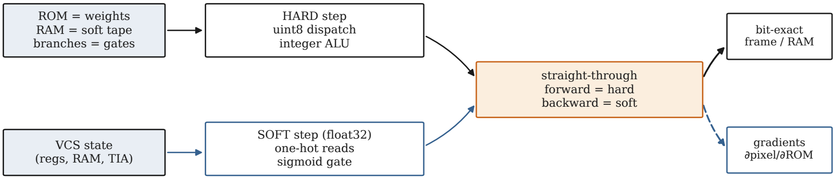
- A real 1977 computer: CPU, RAM, graphics chip, cartridge
- The platform that launched deep reinforcement learning
- Complex, yet fully specified to the last bit
- We re-express its every step so gradients can flow

Two independent differentiable ports

	JUTARI	JAXTARI
Language / AD	Julia / Zygote	JAX / XLA
RAM-exact (of 64)	**64**	**64**
Pixel-exact (of 64)	**64**	**64**
Single env, CPU	fastest	vmap-batched
Batched on GPU	—	$\sim$3M steps/s

- Built twice, in two languages, against one reference
- Both match XITARI bit-for-bit on all 64 games

Soft equals hard



- ROM as a weight tensor, RAM as a soft tape, branches as gates
- Forward pass is bit-exact equal to the hard machine
- Backward pass gives surrogate gradients where bits have none

Theorem 1: exact forward equivalence

For every reachable state x and every finite sharpness α and temperature T,

$$\Phi_{\mathrm{S}}(x) \;=\; \Phi_{\mathrm{H}}(x)$$

in identical float32, so the soft and hard trajectories coincide at every step. The modes differ only in the gradient $\partial\Phi_{\mathrm{S}}/\partial x$, which is defined and nonzero where the hard one is zero or undefined.

Why it matters. Attribution is computed on the *exact* hard trajectory, not an approximation. The forward error is identically zero, at any temperature.

Theorem 2: temperature-limit bound

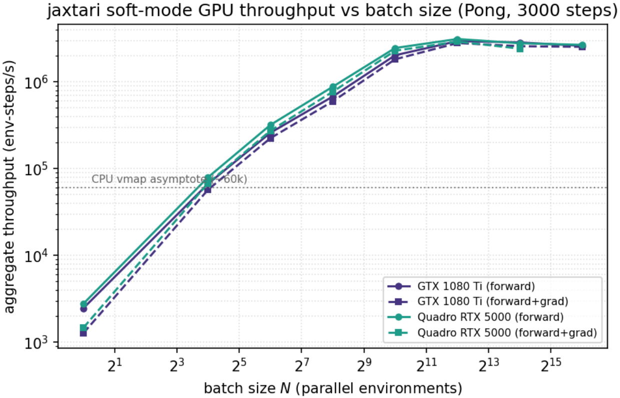
The *fully relaxed* step (softmax dispatch, sigmoid branch, distance-softmax read, *without* the straight-through correction) converges to the hard step as $T \to 0$ and $\alpha \to \infty$:

$$\left\| \Phi_{\mathrm{S}}^{T}(x) - \Phi_{\mathrm{H}}(x) \right\|_{\infty} \leq C \left[(K-1)\, e^{-\Delta_{\mathsf{min}}/T} + e^{-\alpha m_{\mathsf{min}}} + \rho\, e^{-1/T} \right].$$

The deviation falls exponentially as the temperature drops and the sharpness grows.

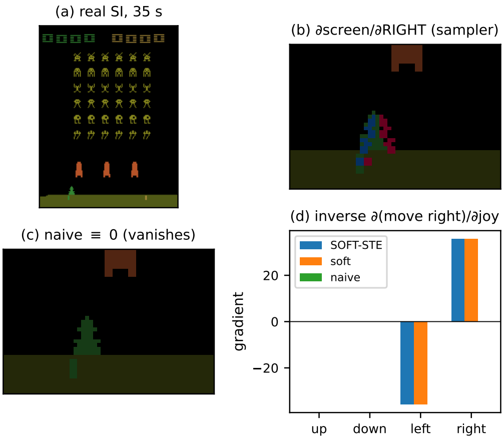
Why it matters. The relaxation is principled even without the straight-through trick. The straight-through estimator reaches this hard limit *exactly* at finite temperature.

Throughput: a GPU path



- JUTARI: fastest per environment on CPU
- JAXTARI: batches the rollout on a GPU
- Millions of environment-steps per second
- Gradients cost only a few percent more

Gradients with a ground truth



- A real Space Invaders ROM, run in the differentiable substrate
- Gradient of the screen with respect to the joystick action
- An attribution that can be scored against the true mechanism

In the supplementary material

- Full proofs of the forward-equivalence and temperature-limit results
- A study of how the relaxation parameters shape the gradient
- Every game, with its ROM hash, mapper, and per-frame exactness
- Throughput tables for CPU and GPU
- These side-by-side comparison videos

A testbed for explainable AI

- A complex system that is fully known and fully differentiable
- Built twice, validated bit-for-bit on all 64 games
- Exact frames, with surrogate gradients for attribution
- The XAI ground truth we hope the community takes up

Full code of both ports released under the MIT license upon acceptance.
Thank you.